

Google Colaboratory & Python

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Access to Colab.

(1) From Google Chrome, go to **Google Drive**.

(2) In the Google Drive, **+ new->others->Google Colaboratory**

guide : <https://colab.research.google.com/notebooks/welcome.ipynb?hl=ja>)

(3) file -> create a new notebook

The screenshot shows a web browser window with two tabs: 'マイドライブ - Google' and 'Untitled0.ipynb - Colab'. The address bar shows the URL 'colab.research.google.com/drive/1zuBM7nCy1TXD_B6jtlmePyyIDtUbya26'. The notebook interface includes a top bar with the 'Untitled0.ipynb' title, a star icon, and a menu with options like 'ファイル', '編集', '表示', '挿入', 'ランタイム', 'ツール', and 'ヘルプ'. Below the menu is a status bar that says 'すべての変更を保存しました'. The main area has a toolbar with '+ コード' and '+ テキスト' buttons. A code cell is visible with a play button icon. Two red callout boxes are present: one pointing to the play button and another pointing to the notebook title.

code cell :
input a program.
execution: push the button or
Control-Enter or Shift-Enter

.ipynb notebook name.
it is created at Google drive ->
MyDrive
-> Colab Notebooks

mount your Drive to Colab.

(1)execute the following commands:

```
> from google.colab import drive  
> drive.mount('/content/drive')
```

MyDrive is mounted at /content/drive

例 Python prog6-0

```
#prog 6-0  A*sin(2*pi*f*t)
```

```
import numpy as np
```

```
import matplotlib.pyplot as plt
```

```
import IPython.display
```

```
print("prog 1-1")
```

```
fs = 8000
```

```
t = np.arange(0,1,1/fs)
```

```
a = 0.8
```

```
f = 440
```

```
y = a*np.sin(2*np.pi*f*t)
```

```
plt.plot(t[0:2000],y[0:2000])
```

```
plt.xlabel("time [s]")
```

```
plt.ylabel("amplitude")
```

```
plt.grid()
```

```
plt.show()
```

```
IPython.display.Audio(y,rate=fs)
```